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“Synopsis”

On

AI powered text and sketch to image and 3D model generator

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MITT



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AI powered text and sketch to image and 3D model generator

Introduction

The fusion of artificial intelligence, machine learning, and deep learning has opened the door to a new era of creativity and design. This project proposes a multimodal system capable of taking text, freehand sketches (captured via camera or drawn on-screen), or images as input and generating AI-driven outputs, including photorealistic images and 3D models. This innovation is especially valuable for industries such as gaming, architecture, product design, and AR/VR.

Overview

The application will accept three types of input:

- 1.Text - Natural language descriptions
- 2.Freehand Sketch - Captured via camera or drawn directly on the app
- 3.Image - Pre-existing photos or illustrations

Based on the input type:

- Text or Sketch Input: The system will generate a high-quality AI image and a corresponding 3D model.
- Image Input: The system will generate a 3D model directly.

The backend utilizes state-of-the-art deep learning models such as diffusion models (e.g., Stable Diffusion, ControlNet), sketch-to-image generators, and 2D-to-3D reconstruction models (e.g., DreamFusion, GET3D).

Problem Statement

Manual 3D modeling is time-consuming and requires specialized skills. Moreover, current systems often handle only a single type of input, limiting creative flexibility. There is a need for an intelligent, multimodal tool that bridges the gap between 2D ideas and 3D realizations in a seamless, user-friendly manner.

Abstract

This project presents a multimodal AI-powered system that transforms freehand sketches, textual descriptions, or images into 3D models. The system uses computer vision for sketch recognition, natural language processing for text interpretation, and deep learning for generating AI images and 3D models. By integrating various neural networks-diffusion models, GANs, and 3D reconstruction algorithms-the app provides a streamlined creative process for designers, educators, and developers.

Drawbacks of Existing Systems

- Limited to single input types (text-only or image-only)
- Often lack accurate 2D-to-3D conversion capabilities
- Require extensive manual input and expertise in 3D modeling tools
- Poor sketch recognition or support for creative, abstract inputs
- Not optimized for mobile or user-interactive environments

Proposed System

- A mobile-friendly app that accepts multimodal input: text, sketch (via drawing or camera), and image
- Uses AI models like CLIP, ControlNet, and diffusion-based generators for sketch-to-image synthesis
- Employs 2D-to-3D reconstruction models like DreamFusion or Zero-1-to-3 for 3D generation
- Seamlessly delivers AI-generated images and/or 3D models depending on the input
- User-friendly UI with real-time preview and export options for 3D models (e.g., OBJ, GLB)

SDG Implemented by this Project

- SDG 9 - Industry, Innovation and Infrastructure: Promotes innovation through cutting-edge AI and DL technologies
- SDG 4 - Quality Education: Enhances learning tools with visual and spatial understanding
- SDG 8 – Decent work and Economic Growth: Boosts productivity in creative Industry

Reference Papers

1. Reality3DSketch: Rapid 3D Modeling of Objects From Single Freehan Sketches Zhang, Y., Xu, Q., Pan, J., et al. (2022). IEEE Transactions on Visualization and Computer Graphics.
<https://ieeexplore.ieee.org/document/9780543>
2. DreamFusion: Text-to-3D using 2D Diffusio Poole, B., Jain, A., Barron, J. T., Mildenhall, B., et al. (2022). Google research
<https://dreamfusion3d.github.io>
3. Text2Mesh: Text-Driven Neural Stylization for Meshes
Michel, O., et al. (2021). NeurIPS 2021.
<https://arxiv.org/abs/2110.01587>
4. 3D-Aware Image Synthesis using 3D Generative Adversarial Networks
Niemeyer, M., Geiger, A. (2021). CVPR.
<https://arxiv.org/abs/2007.06622>